

Javier Huenupi

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I am a second-year Ph.D. Physics student at the University of Pennsylvania, with experience in theoretical physics, particularly in the context of early Universe cosmology.

Education

University of Pennsylvania

Doctor of Philosophy in physics

2025 – present

Universidad de Chile

Master of Science in physics

2023 – 2025

Bachelor of Science in astronomy

2019 – 2023

Publications & Presentations

Papers

- **Huenupi, J.**, Muñoz, C., Palma, G. A., & Sypsas, S. “Pushing the Primordial Frontier: Cosmological Collider Signatures at Strong Mixing,” [arXiv:2607.14529](#) [hep-th]
- **Huenupi, J.**, Muñoz, C., Palma, G. A., & Sypsas, S. “Pushing the Primordial Frontier: Exact Linear Solutions in Multifield Inflation,” [arXiv:2606.18248](#) [hep-th]
- **Huenupi, J.**, Hughes, E., Palma, G. A., & Sypsas, S. “Regularizing infrared divergences in de Sitter spacetime: loops, dimensional regularization and cutoffs,” *Phys. Rev. D*, 110(12), 123536. <https://doi.org/10.1103/PhysRevD.110.123536>
- **Huenupi, J.**, Hughes, E., Palma, G. A., & Sypsas, S. “A note on loop resummation in de Sitter spacetime with the wavefunction of the universe approach,” *Phys. Rev. D* 112, 043543. <https://doi.org/10.1103/nxjc-8qw2>

Talks

- “Regularizing infrared divergences in de Sitter spacetime” - Presented at: [Institut Astrophysique de Paris](#), [Instituto de Física Teórica de Madrid](#), [Universidad Católica de Valparaíso](#), [Universidad Católica](#)
- “Non-Gaussianities from curvature field” - Presented at [Universidad del Bío-Bío](#)

Posters

- “Non-Gaussian distribution of primordial curvature” - Presented at [Graviticulas School](#)
- “Excitation of quantum states during primordial universe” - Presented at [CosmoSur VI conference](#)

Experience

Research

Kinematic flow

August 2025 – present

Advisor: Hayden Lee

Differential equations for correlators and wavefunction coefficients at loop-order in a scalar conformally coupled theory with power-law FLRW cosmology.

Loops resummation and regularization in de Sitter spacetime

October 2022 – December 2024

Advisor: Gonzalo A. Palma and Spyros Sypsas

Computation and regularization of n -point correlation functions with an arbitrary scalar potential.

Bogoliubov coefficients in multi-field inflation

January 2022 – present

Advisor: Gonzalo A. Palma and Claudio Muñoz

Analytical and numerical computation of Bogoliubov coefficients governed by a first-order differential equations system.

Phase transition of higher spin black holes in 2+1-dimensions

March 2022 – February 2023

Advisor: Alberto Faraggi

Analysis of the thermodynamics of $2 + 1$ -dimensional BH using Chern-Simons theory.

Supermassive black hole binary mergers dynamics

January 2022 – April 2022

Advisor: Andrés Escala

Using RAMSES and RAMSES-RT, we ran large-scale 3D simulations and found a significant difference in dust distribution between SMBHBs with and without AGN feedback.

Structures in protoplanetary disks

March 2021 – September 2021

Advisor: Anibal Sierra

Study of dust dynamics in young protoplanetary disks, both analytically and through computer models simulating ALMA observations.

Teaching

Teaching Assistant

2021 – present

Modern Physics, Mechanics (8 times), Electromagnetism, Classical Mechanics (graduate-level), and Quantum Field Theory. I achieved an average grade of 6.7/7.0.

UCH Summer School 2022 Teacher

Astronomy and General Relativity courses for high school students.

Workshops and Schools

I have attended to a total of 14 schools and/or conferences. Some of them are:

- Summer School on Cosmology ICTP, Trieste
- Journeys into Theoretical Physics Perimeter-SAIFR-IFT, Sao Paulo